

DATE UTC

1/1/2021

8/7/2026

CITY

All

STATION NAME

All

QUALITY BAND

All

40.21

Avg PM2.5

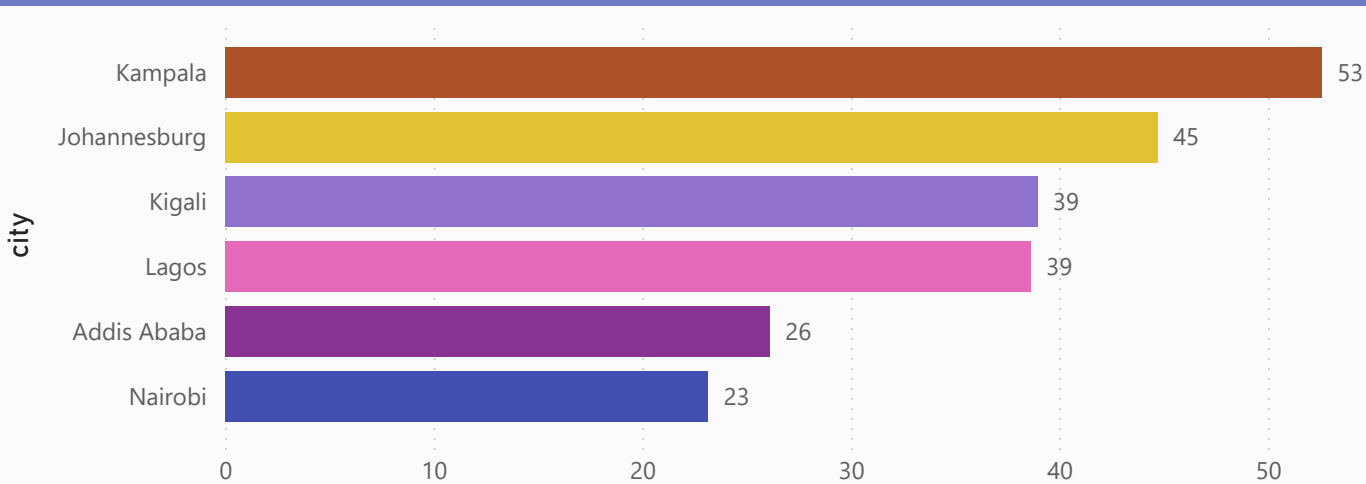
Kampala

Most Polluted City

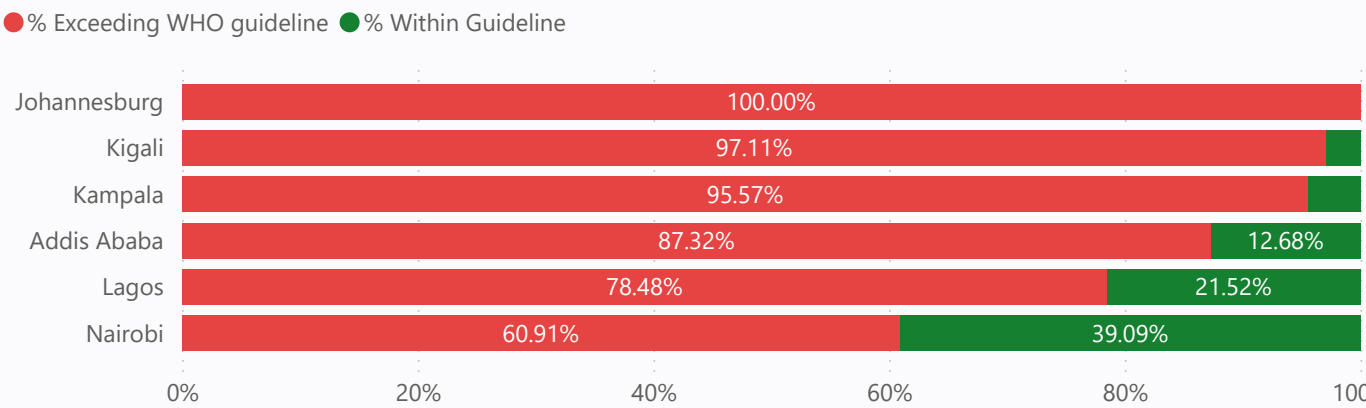
84.48%

% Exceeding WHO guideline

Avg PM2.5 by city



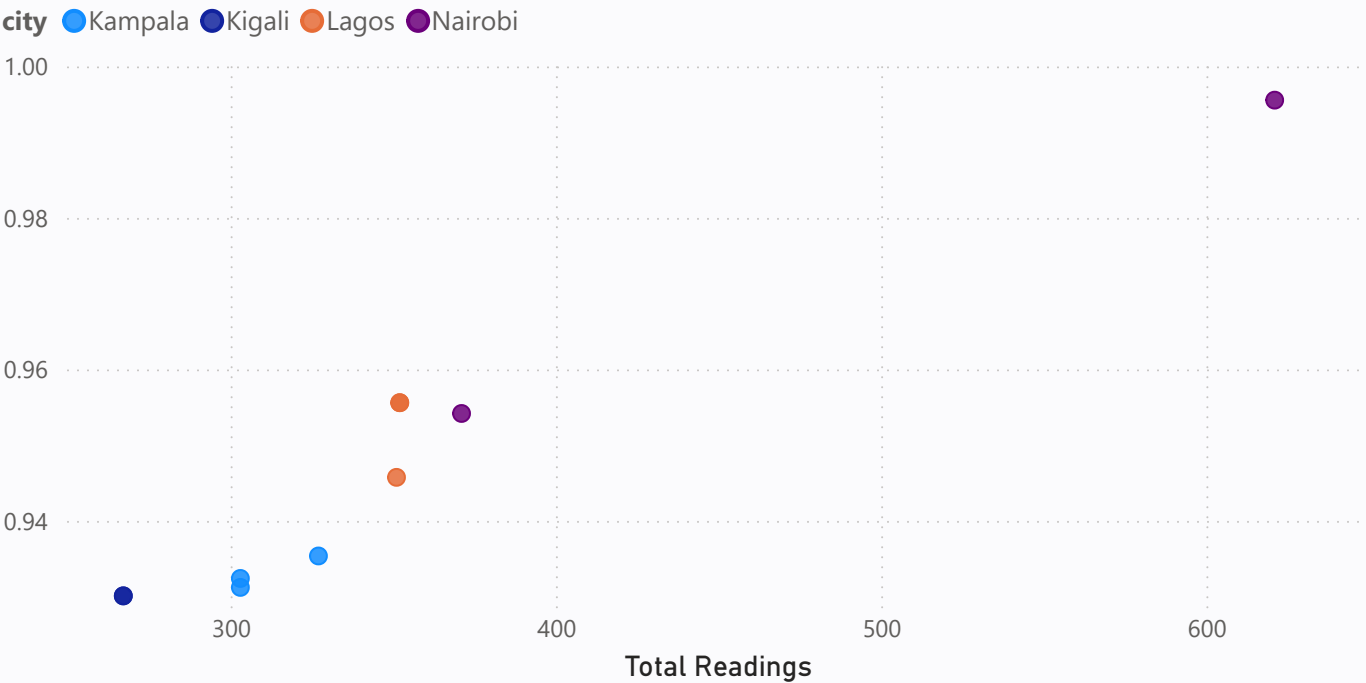
% Exceeding WHO guideline and % Within Guideline by city



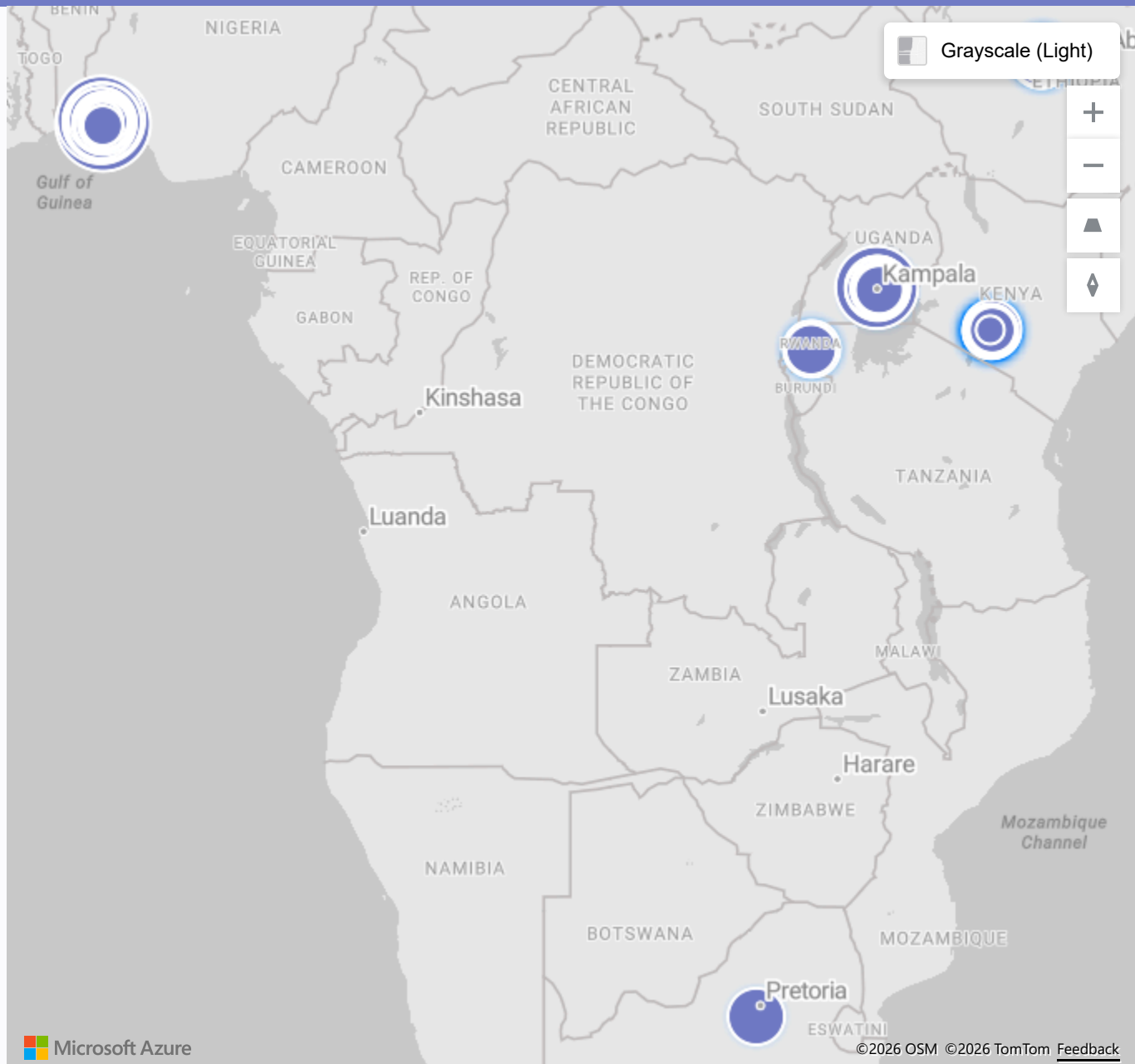
Monthly PM2.5 by City

| city         | January | February | March | April | May   | June  | July  | August | September | October | November |
|--------------|---------|----------|-------|-------|-------|-------|-------|--------|-----------|---------|----------|
| Addis Ababa  | 24.65   | 22.70    | 21.29 | 21.44 | 28.78 | 38.53 | 32.67 | 29.51  | 35.37     | 21.40   | 19.62    |
| Johannesburg |         |          |       |       |       |       |       | 44.70  |           |         |          |
| Kampala      | 66.16   | 53.25    | 48.21 | 35.48 | 40.88 | 48.93 | 56.61 | 53.18  | 51.80     | 62.16   | 61.32    |
| Kigali       | 48.44   | 39.79    | 30.25 | 25.88 | 28.90 | 45.78 | 56.29 | 56.06  | 48.17     | 34.00   | 26.14    |
| Lagos        | 64.27   | 36.15    | 26.46 | 22.58 | 26.23 | 35.34 | 34.08 | 40.75  | 42.44     | 44.13   | 56.12    |
| Nairobi      | 24.23   | 24.41    | 18.31 | 18.34 | 45.84 | 26.21 | 23.82 | 22.74  | 22.99     | 19.76   | 17.07    |
| Total        | 53.40   | 39.96    | 32.43 | 26.43 | 33.10 | 40.60 | 42.49 | 40.81  | 39.50     | 41.13   | 46.84    |

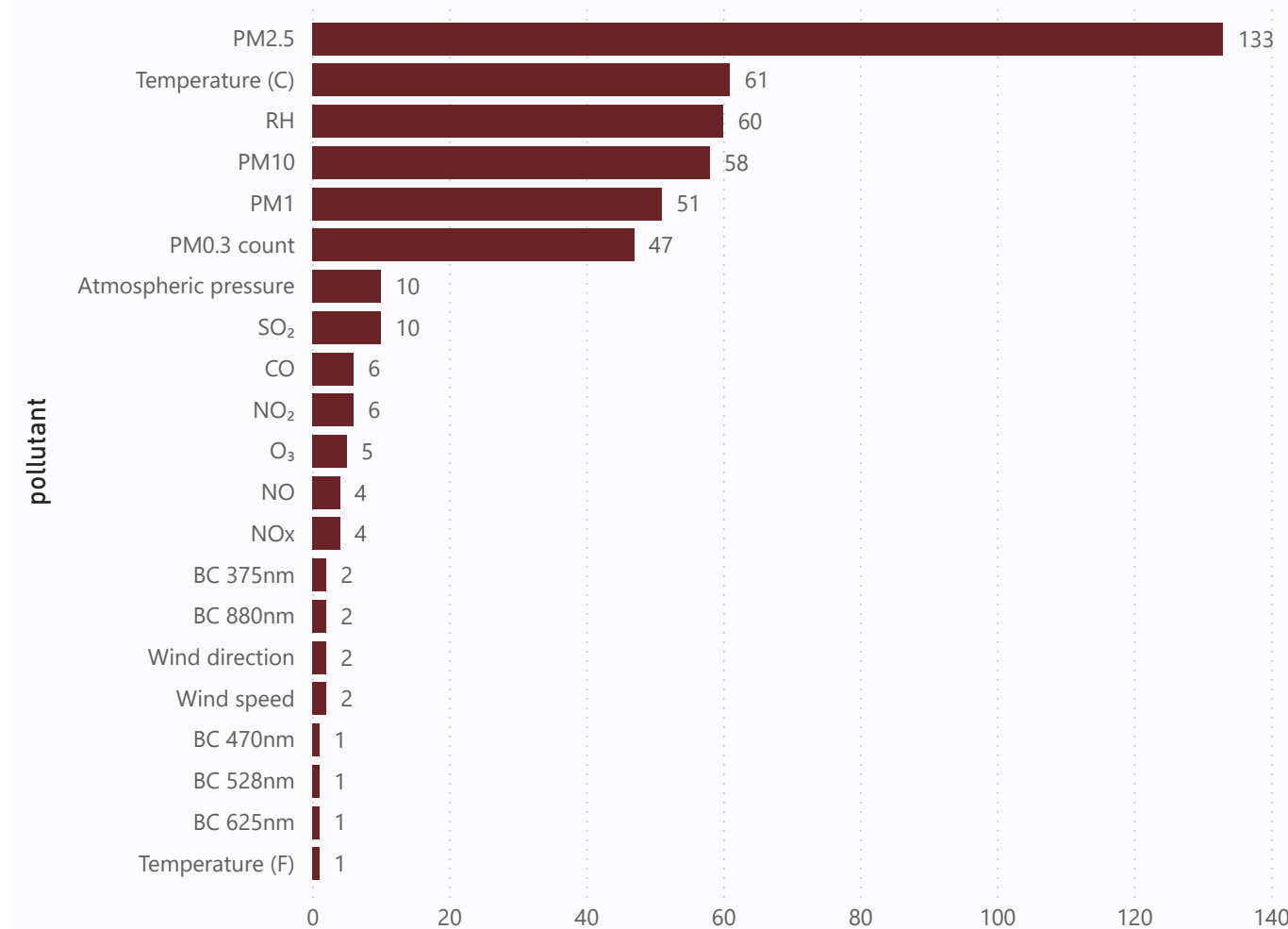
Station Reliability Score by Station



Monitoring stations by PM2.5 level



Station Count by pollutant



Executive summary

Kampala records the highest average PM2.5 concentration among the six cities studied (52.6  $\mu\text{g}/\text{m}^3$ ), followed by Kigali (38.9  $\mu\text{g}/\text{m}^3$ ) and Lagos (38.6  $\mu\text{g}/\text{m}^3$ ). All six cities exceed the WHO annual PM2.5 guideline of 5  $\mu\text{g}/\text{m}^3$ , and among cities with reliable sample sizes, Kigali shows the highest WHO 24-hour guideline exceedance rate at 97.1% of readings, followed closely by Kampala (95.6%) and Addis Ababa (87.3%).

Based on pollution severity, exceedance frequency, and seasonal volatility, **Kampala, Kigali, and Lagos** are identified as the top-priority cities for UNEP intervention. Johannesburg is excluded from this ranking due to insufficient monitoring infrastructure (1 active PM2.5 station, 3 total readings across the study period) and is instead flagged for monitoring-expansion investment rather than direct pollution intervention.